Executive Summary

Aggie Investments Recommendations

Hailey Hendrickson

Introduction

Purpose

- provide informed recommendations regarding the potential of market entry, types of listings to acquire, and how to manage those listings.

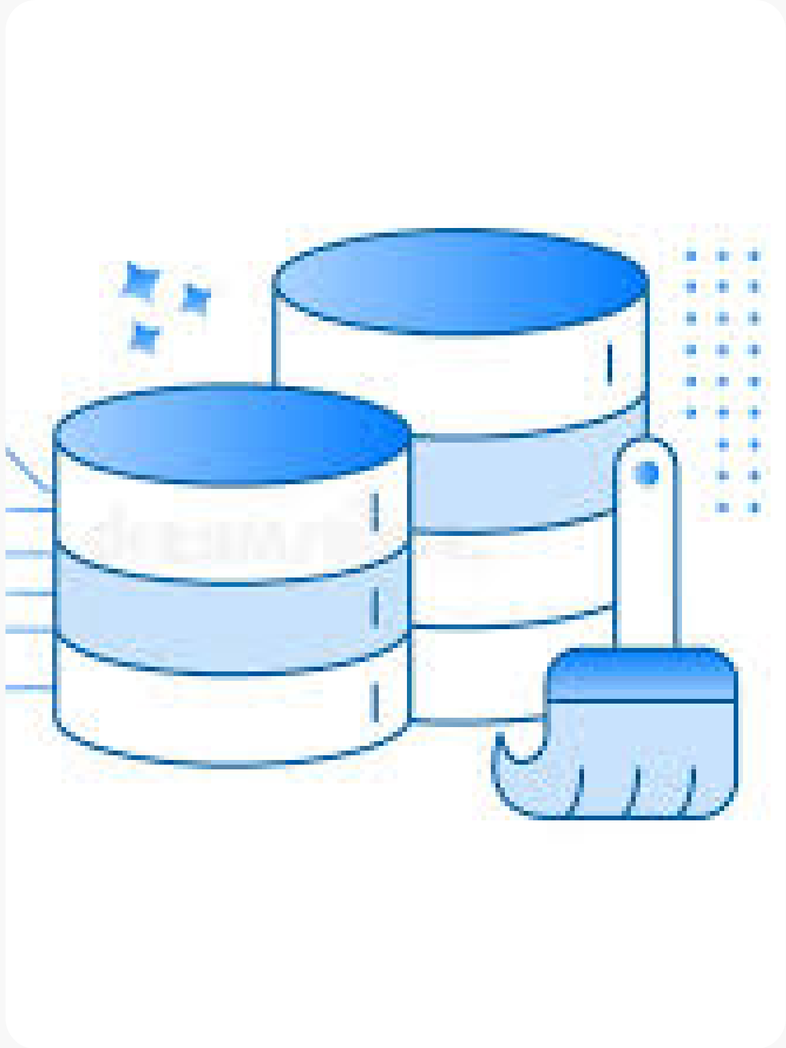
Questions to Investigate

- what kind of listings will provide the most potential for profit?
- what variables are most important to focus on?

Data Information

- Secondary Source
- Contains information about listing options (like duration and count of bedrooms) and scores from guests of AirBnbs.

Methods: Data Preparation



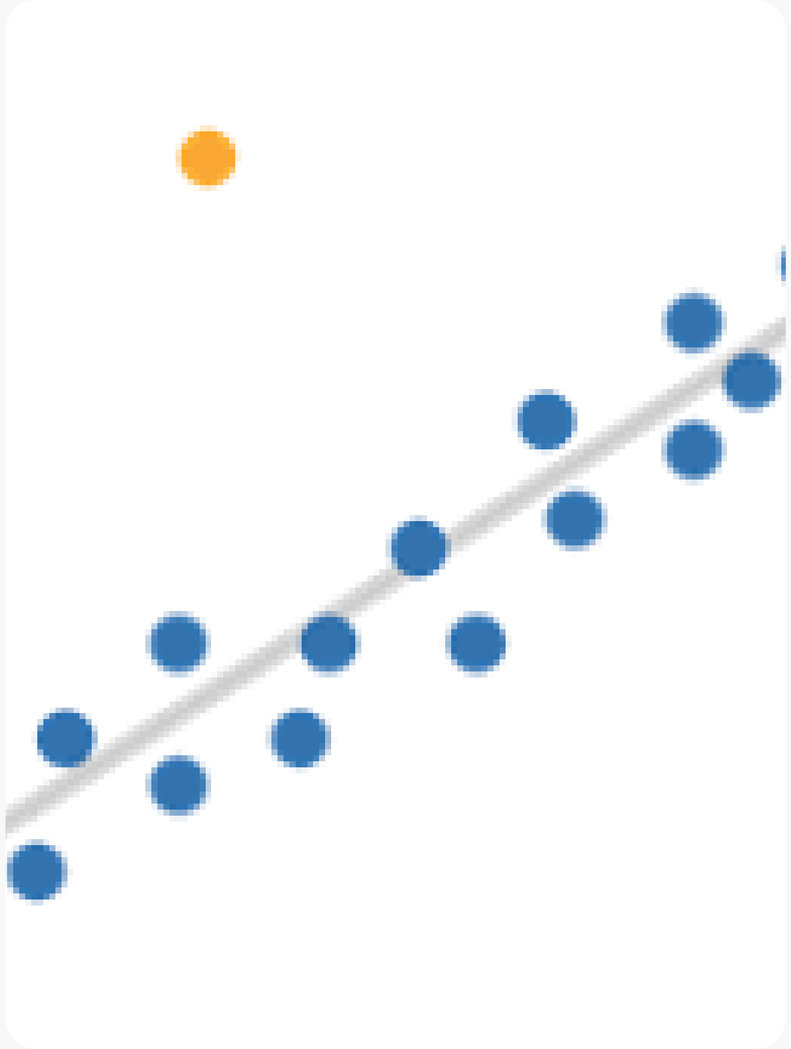
Data Cleansing

- Dropped host name and profile picture to avoid privacy issues
- Changed T/F columns to 1/0
- Condensed room type values for clarity
- Changed bathrooms and price to float data tyles (numerical)

Assumptions and Limitations

- Missing data for air quality and price: certain neighborhood districts had a higher proportion of missing values. Certain neighborhoods could be lower reporting because they don’t want to share poor information

Methods: Data Preparation



Missing Values and Outliers

- Missing data for reviews scores: changed to 0 for the ones that had 0 reviews per month.
- Transformed price and average air quality to a log value to lessen skew of outliers
- Price: performed simple imputation on a log variable
- Avg air quality: performed simple imputation on a log variable
- Bathrooms and bedrooms: found the log to fix outliers.

Methods: Modeling

Spearman Correlation Analysis and T-Tests

- Using a Spearman instead of Pearson correlation because it works better with non-linear data.
- Created a set of hypotheses based on correlation between price and accommodates with a correlation coefficient of 0.557
- 2 sample independent t test
  - Null Hypothesis: There is no difference between price of Airbnbs with hosts that are super hosts and those that are not super hosts.
  - Alternate Hypothesis: AirBnbs with hosts that are super hosts have a higher price than those that don't.
  - p value= 1.691e-24. Because our p value is basically zero, we reject the Null hypothesis. There is significant evidence that AirBnbs with hosts that are super hosts have a higher price than those that don't.

	calculated_host_listings_count	host_listings_count	availability_365	minimum_nights	accommodates	beds	reviews_per_month
calculated_host_listings_count	1.000000	0.606989	0.256190	-0.207885	0.238092	0.101927	-0.073606
host_listings_count	0.606989	1.000000	0.163487	-0.172831	0.082856	-0.083188	-0.271713
availability_365	0.256190	0.163487	1.000000	-0.073053	0.221279	0.169338	-0.011709
minimum_nights	-0.207885	-0.172831	-0.073053	1.000000	-0.243044	-0.102098	-0.415347
accommodates	0.238092	0.082856	0.221279	-0.243044	1.000000	0.355621	0.038029
beds	0.101927	-0.083188	0.169338	-0.102098	0.355621	1.000000	0.215789

Methods: Modeling

Association Rules Analysis

- To examine what types of qualities within an Airbnb property tend to co-occur with good ratings
- Dichotomized necessary variables, generated frequent item sets.
- **Associations of note**:
  - {'short_term', 'accommodates_abovemed', 'bathrooms_abovemed'} → {'price_lg_abovemed'}, **confidence =0.87** , lift = 1.39. This shows that Airbnbs with short term options that have more space for people and bathrooms are usually priced higher, meaning there is more demand.
  - [{'short_term', 'availability_365_abovemed'}], → ['price_lg_abovemed'], **confidence = 0.72** and lift = 1.15. This shows that Airbnbs that have more availability year round but are short term rentals typically are priced higher.
  - [{'availability_365_belowmed', 'review_scores_value_good', 'review_scores_location_good'}], → [{'review_scores_rating_good'}], **confidence =0.99** and lift=1.12. This shows that Airbnbs that are less available can still have a good review rating when they have good location and value.

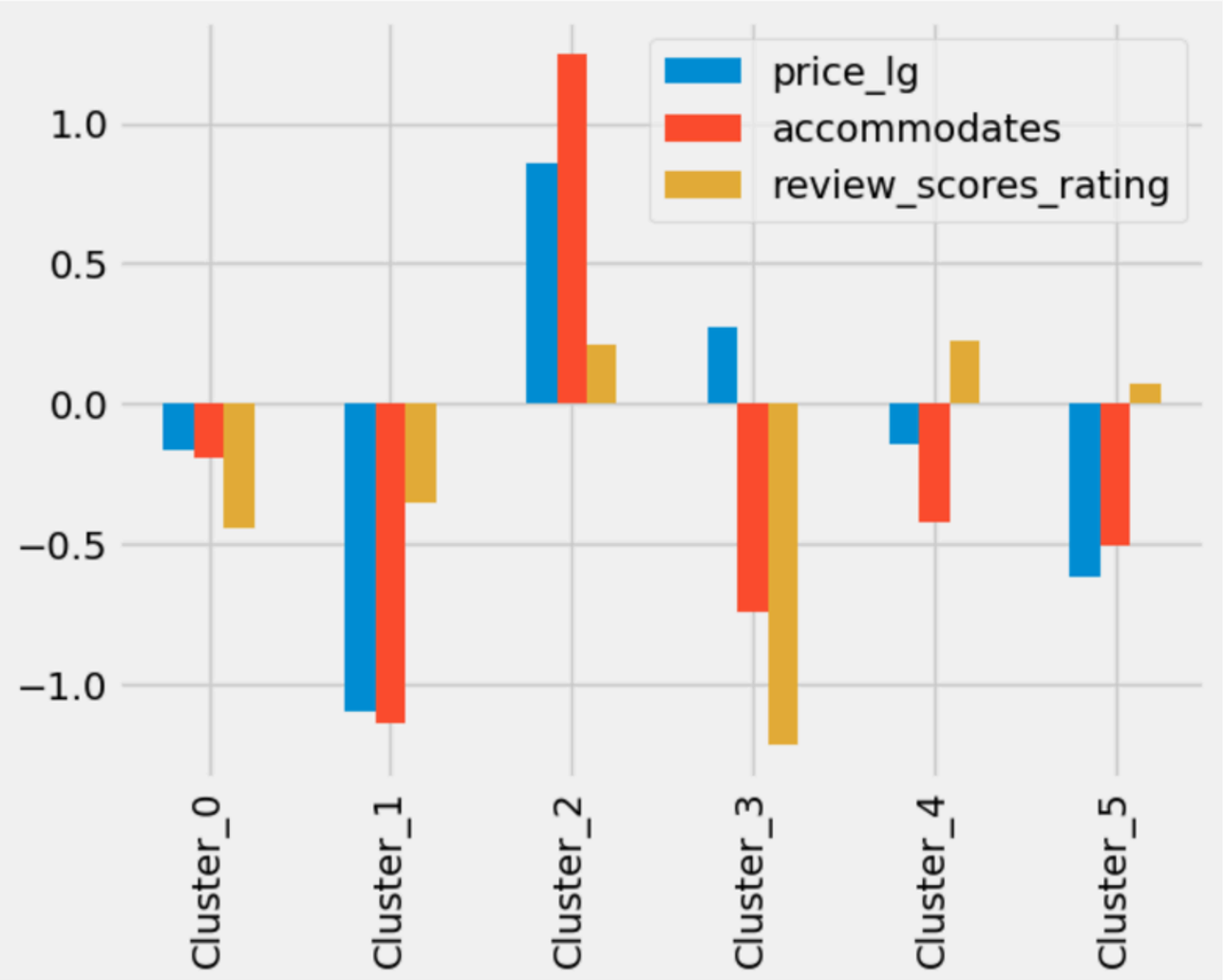
Methods: Modeling

Clustering

- To make final recommendations on segments of properties that Aggie Investments should invest in.
- Dropped redundant and nominal variables, scaled the data
- Used the elbow rule to find kmeans cluster count (6)

RESULTS

- Cluster 2 shows an above average price and accommodates, which a slightly above average review score.
- Cluster 4 shows an above average review score with below average accommodates and slightly below average price.



Results

SuperHost requirement

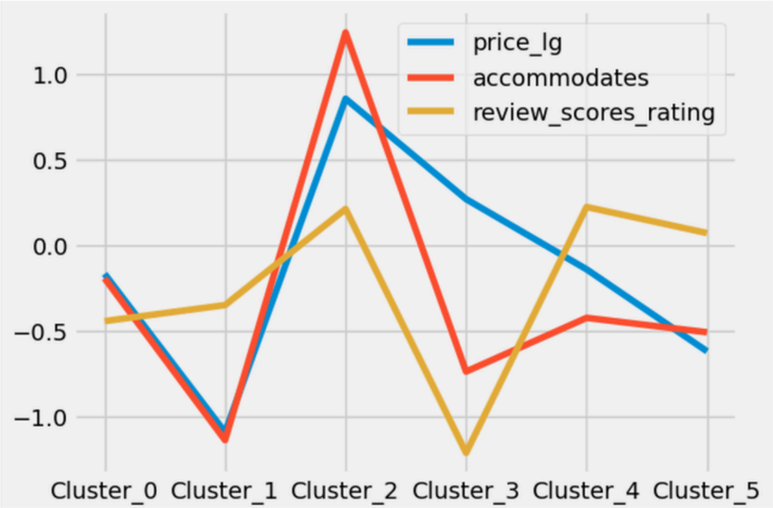
From our T-Test we see that Airbnbs with super hosts typically have a higher price.

Short term and good availability

Airbnbs that have more availability year round but are short term rentals typically are priced higher.

Clusters by price/accommodates

Our clusters typically have correlated price and accommodates.



Recommendations

Host must be super host

- Make for easier management
- Keep a high quality standard

Large Groups

- Short term rentals with high year round availability
- Fit a large number of people

Vacationing Families

- Below average availability, good location (part time rentals?)
- Below average accommodates: smaller groups and families